

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
== RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\num_range.py =
Enter a number: 80
Enter range start: 50
Enter range end: 100
80 is in the range [50,100]
```

```
== RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\num_range.py =
Enter a number: 20
Enter range start: 50
Enter range end: 100
20 is not in the range [50,100]
```

|

Program No - 1**Date:25-07-2025****Aim: Write a program to check whether the number is in range.****Source code**

```
num=int(input("Enter a number: "))
start = int(input("Enter range start: "))
end = int(input("Enter range end: "))
if start<=num<=end :
    print(f"{num} is in the range [{start},{end}]")
else:
    print(f"{num} is not in the range [{start},{end}]")
```

Output

```
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```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\pythagoras.py =
```

```
Enter x1: 1
```

```
Enter y1: 2
```

```
Enter x2: 4
```

```
Enter y2: 6
```

```
The distance between the points (1, 2) and (4, 6) is 5.0
```

Program No - 2

Date: 25-07-2025

Aim: Calculate the distance between two points taking input from the user using Pythagoras theorem.

Source code

```
import math
x1 = int(input("Enter x1: "))
y1 = int(input("Enter y1: "))
x2 = int(input("Enter x2: "))
y2 = int(input("Enter y2: "))
distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
print(f"The distance between the points ({x1}, {y1}) and ({x2}, {y2}) is {distance}")
```

Output

```
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Enter "help" below or click "Help" above for more information.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\library_fine.py
Library Fine System
```

```
Enter the number of days the book is late: 5
Total Fine is Rs 5.
Fine is Rs 1 per day.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\library_fine.py
Library Fine System
```

```
Enter the number of days the book is late: 15
Total Fine is Rs 30.
Fine is Rs 2 per day.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\library_fine.py
Library Fine System
```

```
Enter the number of days the book is late: 25
Total Fine is Rs 125.
Fine is Rs 5 per day.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\library_fine.py
Library Fine System
```

```
Enter the number of days the book is late: 31
Total Fine is Rs 310.
Fine is Rs10 per day.
```

```
Membership is cancelled due to delay of more than 30 days.
```

Program No - 3**Date: 25-07-2025**

Aim: Accept the number of days the member is late to return the book and display the fine or the appropriate message.

Source code

```
print("Library Fine System\n")
days= int(input("Enter the number of days the book is late: "))
if days <= 10:
    fine = 1 * days
    print(f"Total Fine is Rs {fine}.\nFine is Rs 1 per day.")
elif days <= 20:
    fine = 2 * days
    print(f"Total Fine is Rs {fine}.\nFine is Rs 2 per day.")
elif days <= 30:
    fine = 5 * days
    print(f"Total Fine is Rs {fine}.\nFine is Rs 5 per day.")
else:
    fine = 10 * days
    print(f"Total Fine is Rs {fine}.\nFine is Rs10 per day.")
    print("\nMembership is cancelled due to delay of more than 30 days.")
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
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```

```
== RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\fibanocci.py =
```

```
Enter the limit: 10
```

```
0
1
1
2
3
5
8
13
21
34
```


Program No - 4

Date: 25-07-2025

Aim: Write a program to print n terms of Fibonacci series of numbers.

Source code

```
n = int(input(("Enter the limit: ")))  
b1 = 0  
b2 = 1  
f = 0  
for i in range (n):  
    print(f)  
    b1 = b2  
    b2 = f  
    f = b1 + b2
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
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```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\prime_range.py
```

```
Enter start range: 1
```

```
Enter end range: 10
```

```
2
```

```
3
```

```
5
```

```
7
```

Program No - 5

Date: 25-07-2025

Aim: Write a program to find all prime numbers within a given range.

Source code

```
start = int(input("Enter start range: "))
end = int(input("Enter end range: "))
for num in range(start, end + 1):
    if num > 1:
        for i in range(2, num):
            if (num % i) == 0:
                break
        else:
            print(num)
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
== RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\date_time.py =
```

```
Today's date: 2025-07-28
```

```
Current Time: 19:39:03.017994
```

```
|
```

Program No - 6

Date: 28-07-2025

Aim: Print date and time for today and now.

Source code

```
from datetime import datetime
now = datetime.now()
print("Today's date: ",now.date())
print("Current Time: ",now.time())
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\list_function.py
```

```
Enter number of elements: 3
```

```
Enter the elements:
```

```
apple
```

```
orange
```

```
cherry
```

```
['apple', 'orange', 'cherry']
```

```
Enter element to insert: mango
```

```
Enter position to insert: 1
```

```
After insert elements: ['apple', 'mango', 'orange', 'cherry']
```

```
Enter an element to remove: cherry
```

```
After remove : ['apple', 'mango', 'orange']
```

```
Enter a element to append: banana
```

```
Appended list : ['apple', 'mango', 'orange', 'banana']
```

```
Length of the list = 4
```

```
Enter position of element to be popped: 2
```

```
Element after popped: ['apple', 'mango', 'banana']
```

```
Cleared list: []
```

Program No - 7

Date: 28-07-2025

Aim: Create a list and perform insert(), append(), pop(), remove(), len(), clear().

Source code

```
list = []
n = int(input("Enter number of elements: "))
print("\nEnter the elements:")
for i in range(n):
    element = input()
    list.append(element)
print(list)
key = input("\nEnter element to insert: ")
pos = int(input("Enter position to insert: "))
list.insert(pos, key)
print("After insert elements: ",list)
key = input("\nEnter an element to remove: ")
list.remove(key)
print("After remove :",list)
key = input("\nEnter a element to append: ")
list.append(key)
print("Appended list :", list)
print("\nLength of the list =", len(list))
pos = int(input("\nEnter position of element to be popped: "))
list.pop(pos)
print("Element after popped:", list)
list.clear()
print("\nCleared list:", list)
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
= RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\arithmetic_oper.py
```

```
--- Simple Calculator Menu ---
```

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus (a % b)
6. Integer Division (a // b)
7. Exit

```
Enter your choice (1-7): 1
Enter first number: 2
Enter second number: 3
Result of Addition: 2.0 + 3.0 = 5.0
```

```
Enter your choice (1-7): 2
Enter first number: 4
Enter second number: 1
Result of Subtraction: 4.0 - 1.0 = 3.0
```

```
Enter your choice (1-7): 3
Enter first number: 2
Enter second number: 3
Result of Multiplication: 2.0 * 3.0 = 6.0
```

```
Enter your choice (1-7): 4
Enter first number: 8
Enter second number: 4
Result of Division: 8.0 / 4.0 = 2.0
```

```
Enter your choice (1-7): 5
Enter first number: 10
Enter second number: 3
Result of Modulus: 10.0 % 3.0 = 1.0
```

```
Enter your choice (1-7): 6
Enter first number: 30
Enter second number: 3
Result of Integer Division: 30.0 // 3.0 = 10.0
```

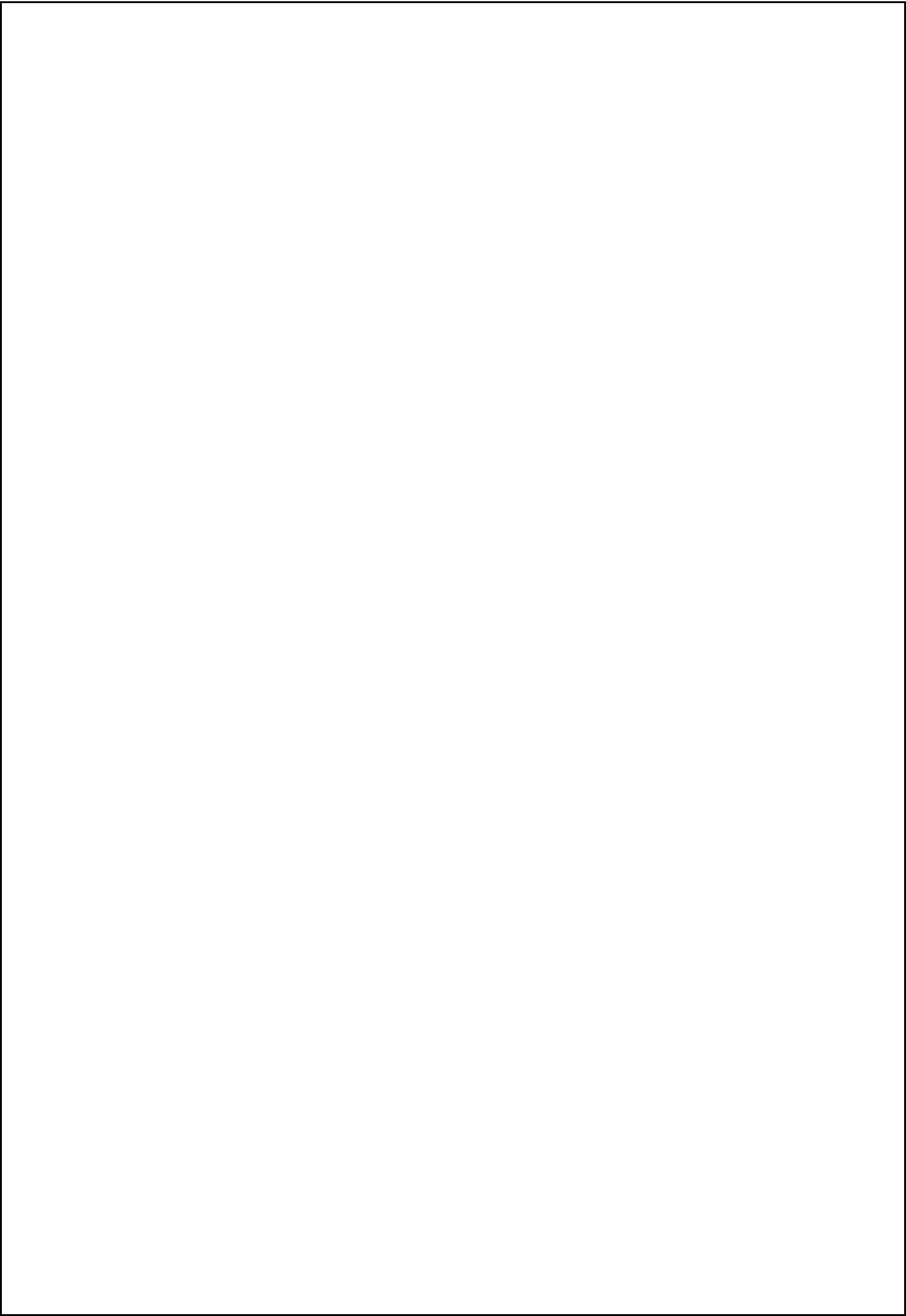
```
Enter your choice (1-7): 7
Exiting calculator. Thankyou
```


Aim: Create a menu with following option

- (i) perform addition**
- (ii) perform subtraction**
- (iii) perform multiplication**
- (iv) perform division**
- (iv) perform modulo division**
- (iv) perform integer division**

Source code

```
print("\n--- Simple Calculator Menu ---")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
print("5. Modulus (a % b)")
print("6. Integer Division (a // b)")
print("7. Exit")
while True:
    choice = input("\nEnter your choice (1-7): ")
    if choice in ['1', '2', '3', '4', '5', '6']:
        try:
            a = float(input("Enter first number: "))
            b = float(input("Enter second number: "))
        except ValueError:
            print("Invalid input. Please enter numbers.")
            continue
    if choice == '1':
        print(f"Result of Addition: {a} + {b} = {a + b}")
    elif choice == '2':
        print(f"Result of Subtraction: {a} - {b} = {a - b}")
    elif choice == '3':
        print(f"Result of Multiplication: {a} * {b} = {a * b}")
```



```
elif choice == '4':  
    if b != 0:  
        print(f"Result of Division: {a} / {b} = {a / b}")  
    else:  
        print("Error: Division by zero.")  
elif choice == '5':  
    if b != 0:  
        print(f"Result of Modulus: {a} % {b} = {a % b}")  
    else:  
        print("Error: Modulus by zero.")  
elif choice == '6':  
    if b != 0:  
        print(f"Result of Integer Division: {a} // {b} = {a // b}")  
    else:  
        print("Error: Integer division by zero.")  
elif choice == '7':  
    print("Exiting calculator. Thankyou")  
    break  
else:  
    print("Invalid choice. Please enter a number from 1 to 7.")
```

Output

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
== RESTART: D:\Akshay MCA Docs Main\MCA\SEM 3\Python lab programs\factorial.py =
Enter the number:5
Factorial of 5 is 120
|
```

Program No - 9

Date: 28-07-2025

Aim: Find the factorial of a number using function.

Source code

```
def factorial(n):  
    if n==0 or n==1:  
        return 1  
    else:  
        return n * factorial(n-1)  
num=int(input("Enter the number:"))  
print(f"Factorial of {num} is",factorial(num))
```